ISA AND CORE2350B (RP2350B)

SAMPLE RAM: on the module

The Core2350B2 has 8 MB of PSRAM on the RP2350B's QMI (CS1 = GPIO47, memory mapped). It replaces v2.0's U8 (APS6404L on GPIO0–3, PIO SPI); GPIO0–3 are now free.

IRQ/DMA SETTING

I2S DAC

POWER

ADDR/DATA MUX/DEMUX

MISC LEVEL SHIFT/BUFFER

AEN IO MASKING

Disables IOR and IOW signals when other DMA in system happens

RESET CIRCUIT

Latches BUSOE high after ADS is toggled for the first time, sets low on RESET

I/OCHRDY OPEN DRAIN

MOUNTING

74AHCxx LOGIC PWR

DACK INVERT

Saves PIO instructions due to JMP limitations

DECOUPLING CAPS

MIDI/DEBUG OUT

USB-C PROGRAM PORT

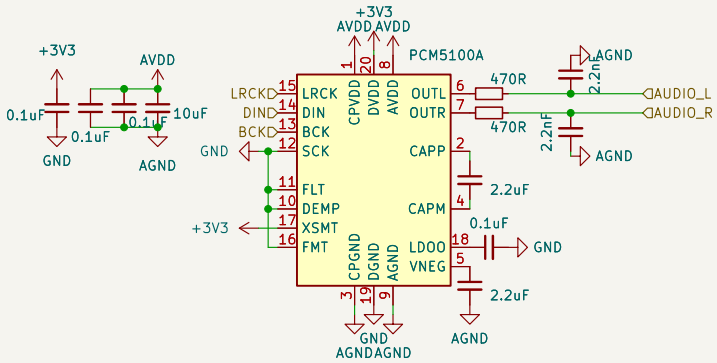
U6 74AHCT126 on +5V: OE high pulls the output low, OE low leaves it open (replaces the 74LVC2G06). Spare gates: A and OE tied low.

U12 module power: VSYS feeds the module's VBUS pin and its own LDO (ME6217, 3V3_EN pulled up on the module) powers the module. The module 3V3 pin is not connected: U9 powers the card logic, DAC and PSRAM. GPIO30–47 are not used (GPIO39: module LED, GPIO47: module PSRAM CS on Core2350B1/B2).

5.1k Rd on CC1/CC2: device only, never sources VBUS. D+/D− shared with J5 (USB host) as on v2.0.

v2.0 ISA logic and GPIO map; TSSOP/SOIC-only chips as SMD
Through-hole PicoGUS 2.0 on a Waveshare Core2350B (RP2350B) module

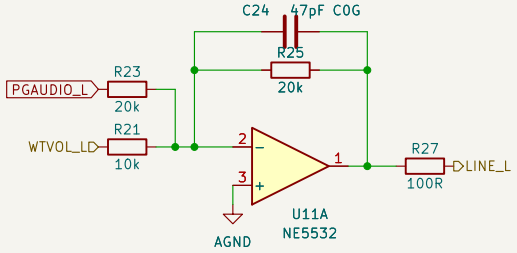
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Title: PicoGUS v2.1 THT		
Size: A3	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 1/3

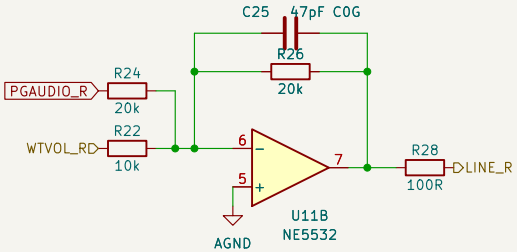


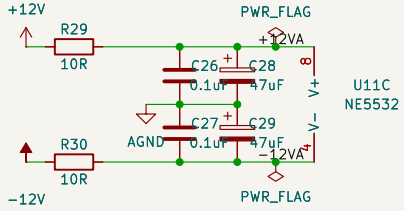
Sheet: /Audio/		
File: audio.kicad_sch		
Title: PicoGUS v2.1 THT – PCM5100A DAC		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 2/3

LINE-OUT MIXER

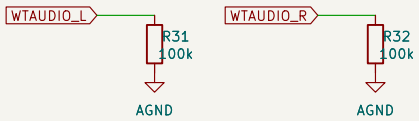
Inverting summing amplifier from THT v1.3: Waveblaster (after U13 M62429 volume) x2 = +6 dB through R21/R22 10k.
PCM5100A x1 through R23/R24 20k, feedback 20k || 47 pF (pole ~170 kHz). U11 runs from ISA +/-12V through 10R / 47uF.
The DAC output is ground-centred, so the PCM path is DC coupled (no electrolytic); the M62429 output sits at about
VCC/2, so C41/C42 (film) couple it. Both channels invert, so left and right keep their relative phase.







WT input pull-downs (THT v1.3 R14/R15)



Sheet: /Mixer/		
File: mixer.kicad_sch		
Title: PicoGUS v2.1 THT – line-out mixer		
Size: A4	Date: 2026-10-29	**Rev: v2.1**
KiCad E.D.A. 10.0.7		Id: 3/3